

Tableau Basics Assignment

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

Answer:

Tableau is a **business intelligence and data visualization tool** that allows users to connect to various data sources, analyze data, and create interactive visual reports and dashboards — without requiring any programming or coding knowledge. It was developed by Tableau Software and is now owned by Salesforce.

Tableau works on a simple drag-and-drop interface where users can build charts, graphs, maps, and dashboards within minutes by simply dragging data fields onto a canvas.

Why Tableau is Preferred Over Traditional Spreadsheet Tools

1. Handles Large Volumes of Data

Traditional spreadsheet tools like Excel have row limitations and slow down significantly with large datasets. Tableau can handle millions of rows of data efficiently without any performance issues.

Example: Excel struggles with a dataset of 2 million sales records while Tableau processes and visualizes the same data in seconds.

2. Interactive and Dynamic Visualizations

Spreadsheets produce static charts that do not respond to user interaction. Tableau creates fully interactive dashboards where users can click, filter, drill down, and explore data in real time.

Example: A Tableau dashboard allows a sales manager to click on a specific region and instantly see only that region's sales performance — without rebuilding the chart.

3. Connects to Multiple Data Sources

Excel is limited to manual file imports. Tableau can connect to databases, cloud storage, APIs, Google Sheets, SQL servers, and many other sources simultaneously — combining data from all of them in one view.

4. Real-Time Data Updates

Tableau dashboards can be connected to live data sources and refresh automatically — ensuring users always see the most current data. Spreadsheets require manual updates every time the data changes.

5. No Coding Required

Unlike traditional data analysis tools that require SQL or Python knowledge, Tableau's drag-and-drop interface allows business users with no technical background to build powerful dashboards independently.

6. Advanced Visualizations

Tableau offers a wide range of visualization types — heat maps, tree maps, geographic maps, scatter plots, and waterfall charts — that are not available in standard spreadsheet tools.

Two Industries Where Tableau is Commonly Used

Industry 1: Healthcare

Tableau is widely used in hospitals, pharmaceutical companies, and health departments to track patient outcomes, monitor disease spread, analyze treatment effectiveness, and manage hospital resources.

Use Case: During the COVID-19 pandemic, health departments used Tableau dashboards to track daily case counts, vaccination rates, and hospital bed availability across different regions — enabling governments to make fast and informed decisions on resource allocation and lockdown policies.

Industry 2: Retail and E-Commerce

Retail companies use Tableau to analyze sales performance, monitor inventory levels, understand customer buying behavior, and measure the effectiveness of marketing campaigns across different channels and regions.

Use Case: A large retail chain uses a Tableau dashboard to monitor daily sales across 500 stores nationwide. Store managers can filter by product category, region, or time period to identify underperforming stores and take corrective action before the end of the month.

Q2. After connecting to the Sample – Superstore dataset:

- a) List the main types of fields available (examples: time-based, geographical, numerical).
- b) Identify any four Dimensions and any four Measures from the dataset.

Hint: Look at the Data Pane on the left side of Tableau and observe blue (Dimensions) and green (Measures) fields.

Answer:

In Tableau, when a dataset is loaded, all fields appear in the **Data Pane** on the left side of the screen. These fields are automatically classified into two categories **Dimensions** (shown in Blue) and **Measures** (shown in Green) based on their data type and role in analysis.

a) Main Types of Fields Available in Sample Superstore

After connecting to the Sample Superstore dataset and observing the Data Pane, the following main types of fields are available:

1. Time-Based Fields Fields that represent dates and time periods. Used for trend analysis and time series visualizations.

Examples: Order Date, Ship Date

2. Geographical Fields Fields that represent physical locations. Tableau automatically recognizes these and assigns a globe icon. Used for map-based visualizations.

Examples: Country, State, City, Region, Postal Code

3. Categorical Fields Fields that represent text-based categories used to group and segment data.

Examples: Category, Sub-Category, Segment, Ship Mode, Customer Name

4. Numerical Fields (Measures) Fields that contain quantitative values that can be aggregated — summed, averaged, or counted.

Examples: Sales, Profit, Quantity, Discount

b) Four Dimensions and Four Measures

Four Dimensions (Blue Fields):

Dimension	Description
Category	Type of product — Furniture, Technology, Office Supplies
Region	Geographical sales region — East, West, North, South
Order Date	Date when the customer placed the order
Sub-Category	Specific product type within a category — Chairs, Phones, Binders

Four Measures (Green Fields):

Measure	Description
Sales	Total revenue generated from orders
Profit	Net profit earned after deducting costs
Quantity	Number of product units sold per order
Discount	Percentage of discount applied to an order

Q3. Create a Bar Chart showing Total Sales by Category.

Explain:

Which field goes to Rows

Which field goes to Columns

Which aggregation is applied to Sales

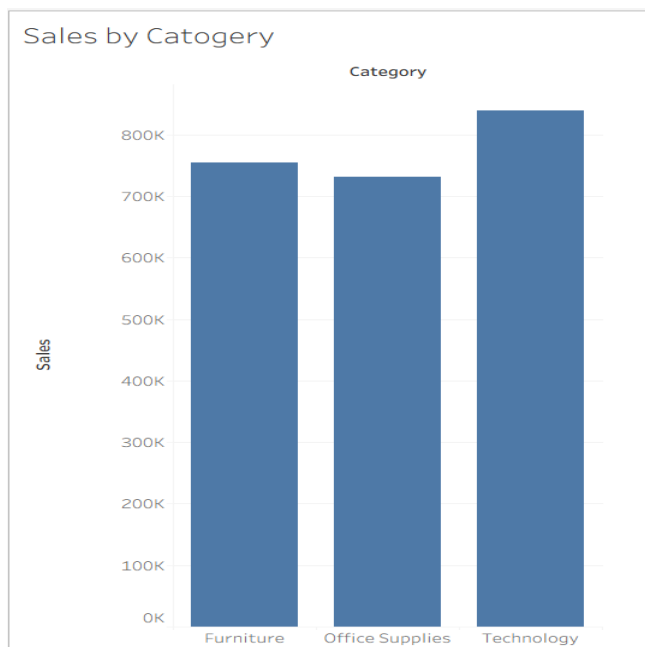
Hint: Drag Category and Sales into the view and observe Tableau's default behaviour.

Answer:

In Rows there will be **Sales**

In Columns There will be Product **Category** field

Tableau default output is **Bar Graph**.



Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category.

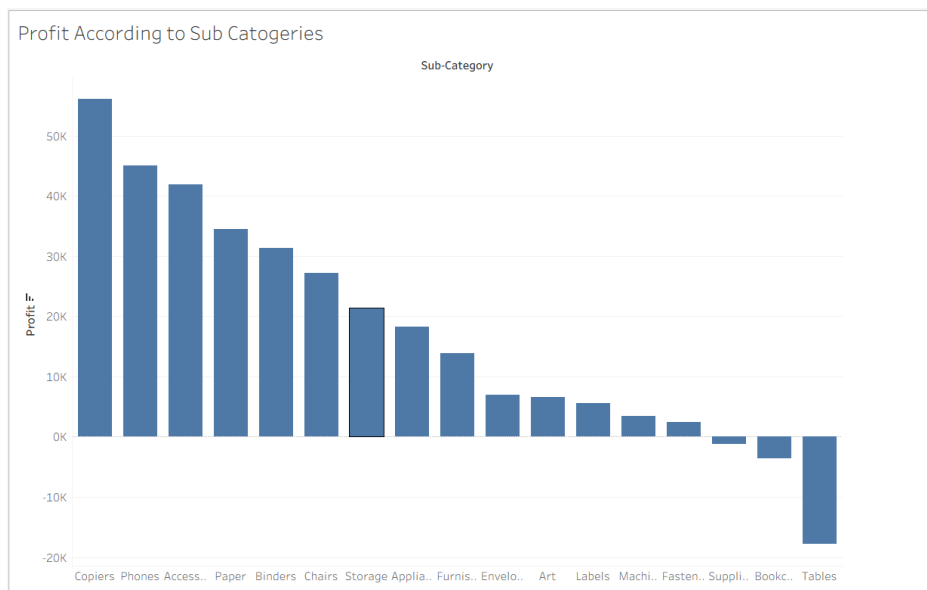
Hint: Use Sub-Category as a Dimension and Profit as a Measure

Answer:

Comparing Sales and Profit together tells you that **high sales doesn't always mean high profit**. For example:

- A Sub-Category might have very high Sales but very low (or even negative) Profit — meaning the business is spending too much to sell it.
- Another might have moderate Sales but high Profit — meaning it's very efficient.

In short, **Sales = Revenue earned**, **Profit = What actually stays after costs**. Together they reveal which products are truly performing vs. just appearing to perform.



Q5. Create a Line Chart to show Sales Trend over Order Date.

Which date level would you choose (Year / Quarter / Month)?

Why are line charts suitable for time-based analysis?

Hint: Drag Order Date and change its date hierarchy from Year to Month if needed

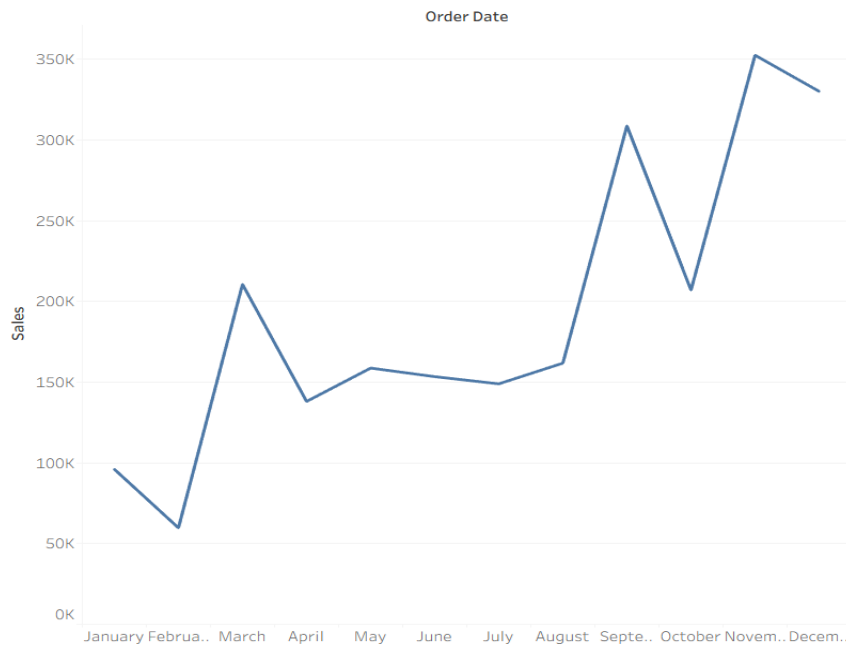
Answer:

Line charts are best for time-based data because:

- They show **continuity** — connecting data points with a line makes trends, rises, and falls easy to spot visually.
- They clearly show **patterns over time** like seasonality, growth, or decline.
- They are specifically designed to answer "**How did this change over time?**"

Bar charts show comparison, Pie charts show proportion — but **Line charts show change over time**. That's why they're the go-to for trends.

Sales According to Months



Q6. Apply a Basic Filter to display data only for:

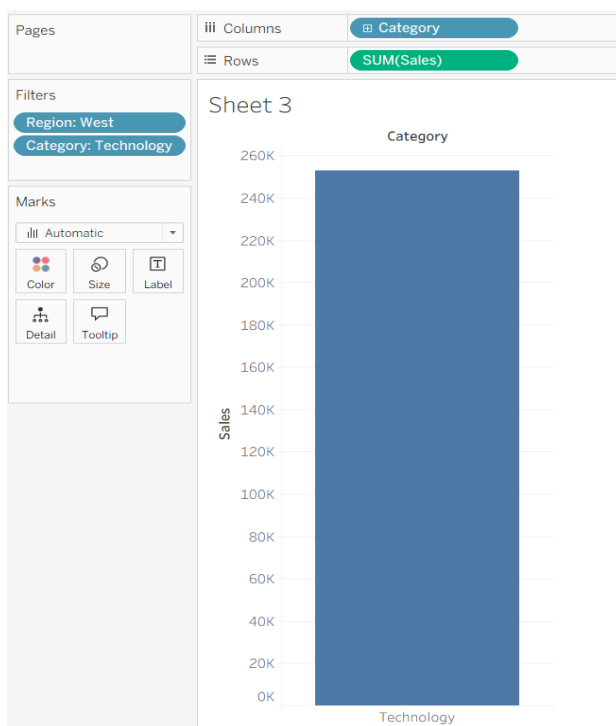
Region = “West”

Category = “Technology”

Explain the steps and mention which shelf is used.

Hint: Drag the required fields to the Filters shelf and select appropriate values.

Answer:



Q7. Explain the business importance of geographical analysis. Then describe how you would create a Map showing Sales by State using Sample – Superstore.

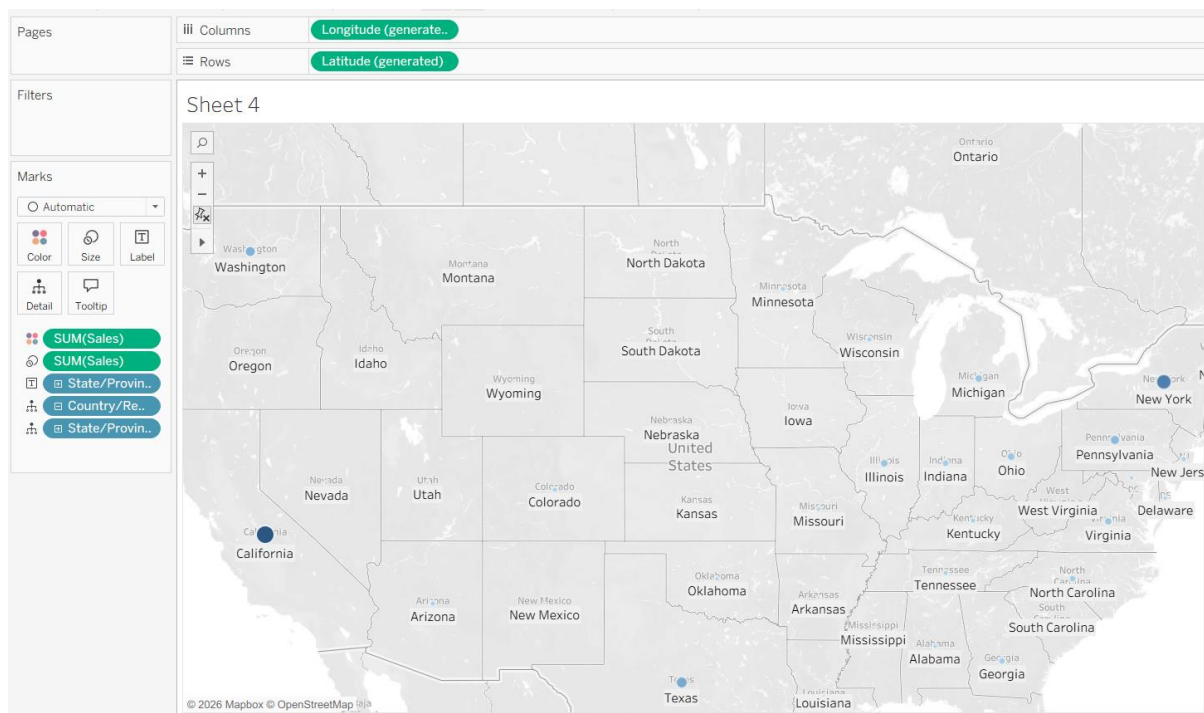
Hint: Use State and Sales. Tableau automatically detects geographic fields

Answer:

Geographical analysis helps businesses understand **where** things are happening, not just what is happening. For example:

- **Which states/regions generate the most revenue?** → Focus sales efforts there.
- **Which areas have low sales?** → Opportunity to expand or investigate.
- **Where are losses happening?** → Cut costs or restructure in those regions.
- **Logistics & Supply Chain** → Understanding geography helps optimize delivery routes and warehouse locations.

In short, geography answers the business question "**Where is our business strong and where is it weak?**" which is impossible to see from just a table or bar chart.



Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

Answer:

What is a Worksheet?

A Worksheet is a **single view/chart** that you build in Tableau. Every visualization you create a bar chart, line chart, map lives on its own individual worksheet.

- One worksheet = One visualization
- It is where all the **building and analysis** happens
- Think of it as a **single page** with one chart on it

What is a Dashboard?

A Dashboard is a **collection of multiple worksheets** combined together on one single screen.

- It pulls together different charts/maps/tables into **one unified view**
- You don't build charts on a dashboard you just **arrange already built worksheets** onto it
- Think of it as a **notice board** where you pin multiple charts together

Key Differences

Point	Worksheet	Dashboard
Contains	Single chart/view	Multiple charts/views
Purpose	Building & exploring data	Presenting & monitoring
Audience	Analyst building the report	Manager/Decision maker viewing it
Interactivity	Limited	High filters on one chart affect others
Created how	Drag & drop fields	Drag & drop worksheets

Q9. What is Tableau Prep, and why is it important in real-world data projects?

What is Tableau Prep, and why is it important in real-world data projects?

Then explain how you would:

Remove null values

Remove duplicate records

Change incorrect data types (Answer in steps, no tool access required.)

Answer:

What is Tableau Prep?

Tableau Prep is a **data cleaning and preparation tool** by Tableau. Before you can analyze data in Tableau Desktop, your data needs to be **clean, correct and structured** that's exactly what Tableau Prep does.

Think of it this way:

- **Tableau Desktop** = Where you analyze and visualize data
- **Tableau Prep** = Where you clean and prepare data before analysis

In real world projects, **data is never perfect** it comes with missing values, duplicates, wrong data types, inconsistent formatting etc. Tableau Prep fixes all of that before it reaches Tableau Desktop.

Why is it Important in Real World Projects?

In real world scenarios:

- Data comes from **multiple sources** Excel, databases, CRM systems etc.
- It contains **errors, nulls, duplicates** that can give wrong insights
- A business decision based on **dirty data** = **wrong decision**

The golden rule in data analysis is:

"Garbage In = Garbage Out"

If you put bad data into Tableau Desktop, your charts and dashboards will show incorrect results no matter how good they look. Tableau Prep ensures **clean data goes in = accurate insights come out**.

How to Remove Null Values (Steps)

Null means **empty/missing value** in a field.

1. Open **Tableau Prep Builder** and connect your data source.
2. Add a **Clean Step** by clicking the "+" button on the flow and selecting "**Clean Step**".
3. In the Clean Step, look at the field/column that has null values you'll see a **grey bar at the bottom** of the field distribution representing nulls.
4. Click on that **grey null bar** to select all null rows.
5. Right-click and choose "**Exclude**" to remove those rows from the dataset.
6. Nulls are now removed from your flow.

How to Remove Duplicate Records (Steps)

Duplicates are **repeated rows** that show the same data more than once.

1. In Tableau Prep, add a **Clean Step** to your flow.
2. Go to the top menu inside the Clean Step and look for the option "**Remove Duplicates**" in newer versions it's under the "**More Options / Data Quality**" menu.

3. Alternatively, you can add an **Aggregate Step** by clicking "+" → "**Aggregate**".
4. Group by all the key fields this automatically collapses duplicate rows into one.
5. This ensures each record appears **only once** in your output.

How to Change Incorrect Data Types (Steps)

Data type mismatch is very common for example a **date stored as text** or a **number stored as string**.

1. In Tableau Prep, open your **Clean Step**.
2. At the top of each column/field you'll see a **data type icon**:
 - **Abc** = Text/String
 - **#** = Number
 - **Calendar icon** = Date
 - **True/False** = Boolean
3. If a field has the **wrong data type** for example Order Date showing as "Abc" instead of a calendar icon click on that **icon**.
4. A small dropdown appears with all available types select the **correct type** (e.g. Date, Number, String).
5. Tableau Prep will automatically **convert the field** to the correct type.
6. If conversion fails for some values, Tableau Prep will flag them as **errors in red** so you can fix them manually.

Q10. A retail manager wants answers to the following:

Which category generates the highest profit?

Which region performs poorly?

How does sales change over time?

- a) Mention which charts you would use
- b) Identify one key insight that management could derive
- c) Explain how this helps in business decision-making

Answer:

Which Category generates the highest Profit?

a) Chart to use Bar Chart

- Drag **Category** to Columns and **Profit** to Rows

- Sort in descending order to clearly see which category is on top

b) Key Insight:

- **Technology** generates the highest profit while **Furniture** may show very low or even negative profit despite having decent sales

c) Business Decision:

- The manager should **invest more in Technology** increase stock, run promotions
- For Furniture **review pricing, supplier costs or discounts** being offered since it sells but doesn't profit well
- Simple rule → **Double down on what works, fix what doesn't**

Which Region performs poorly?

a) Chart to use — Bar Chart or Map

- Bar Chart → Drag **Region** to Columns and **Profit** to Rows for direct comparison
- Map → Drag **State** and color by **Profit** to see geographic performance visually

b) Key Insight:

- The **Central region** typically shows the weakest profit performance in Superstore data sometimes even running at a loss

c) Business Decision:

- Manager can investigate **why Central is underperforming** is it high discounts? High shipping costs? Wrong product mix?
- Could decide to **reallocate budget and sales team efforts** from Central to stronger regions like West or East
- Or launch a **targeted campaign specifically for Central** to turn it around

How does Sales change over Time?

a) Chart to use Line Chart

- Drag **Order Date** to Columns (set to Month level) and **Sales** to Rows
- Line chart automatically shows the trend over time

b) Key Insight:

- Sales consistently **peak towards end of year** (October to December) and **dip at the start of year** (January to February) showing clear seasonality

c) Business Decision:

- Manager should **stock up inventory before October** to meet holiday demand
- Run **discount campaigns in January/February** to boost slow period sales
- Plan **staff hiring and logistics** around peak months to avoid delays

